

Polynomial Functions Quiz

Algebra 2 Version A

Name: _____

1. (4 pts) Factor completely: $2x^2 - 2x - 12$

Answer: $2(x - 3)(x + 2)$

2. (8 pts) Let $f(x) = x^2 + 3x - 4$. Use algebra to answer the following.

a. Find all real zeros of f .

$-4, 1$

b. Find the y -intercept.

$(0, -4)$

c. Describe the end behavior of f .

$f(x) \rightarrow +\infty$ as $x \rightarrow \pm\infty$

d. State the interval(s) where $f(x) < 0$.

$(-4, 1)$

3. (10 pts) Let $f(x) = x^3 - 5x^2 + 2x + 8$. Use algebra to answer the following.

a. Find all real zeros of f .

$-1, 2, 4$

b. Find the y -intercept.

$(0, 8)$

c. Describe the end behavior of f .

$f(x) \rightarrow -\infty$ as $x \rightarrow -\infty$, $f(x) \rightarrow +\infty$ as $x \rightarrow +\infty$

d. State the interval(s) where $f(x) > 0$.

$(-1, 2) \cup (4, +\infty)$

e. What is the maximum number of turning points f can have?

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4. (12 pts) Let $f(x) = 8 - 2(x - 3)^2$, written in vertex form $a(x - h)^2 + k$.

a. Describe the three transformations applied to $y = x^2$ to produce f .

i. Horizontal shift: right 3 units.

Replace x with $x - 3$; shifts the graph right 3.

ii. Vertical stretch/reflection by a factor of 2 with a reflection over the x -axis.

Multiply output by -2 ; stretches by 2 and flips.

iii. Vertical shift up 8 units.

Add 8 to the output; shifts the graph up 8.

b. Find the key features of f . Note how sub-parts i–iv are arranged in a two-column grid.

i. State the vertex.

$(3, 8)$

ii. State the axis of symmetry.

$x = 3$

iii. Find the x -intercepts algebraically.

$x = 1, 5$

iv. Find the y -intercept.

$(0, -10)$

c. State the range of f using interval notation.

$(-\infty, 8]$